Simulation Report

Topic:

Pipeline Processor Design and Simulation

Internship Task 3 – VLSI Design

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Tool Used: Cadence Genus, SimVision

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Chapter 1
Introduction

1.1 Introduction

Digital signal processing systems often require filtering techniques to remove unwanted noise and improve signal quality. A digital filter processes discrete-time signals and produces a modified output signal based on specific filtering requirements. Among various digital filters, the **Finite Impulse Response (FIR) filter** is widely used due to its stability, linear phase response, and simple implementation.

An FIR filter generates an output signal based on the weighted sum of the current input sample and several previous input samples. The weights used in the computation are called **filter coefficients**, and they determine the characteristics of the filter such as low-pass, high-pass, or band-pass filtering.

In this project, a digital FIR **filter** is designed using Verilog Hardware Description Language (HDL) and verified using Cadence simulation tools. The functional behavior of the filter is tested using a testbench, and simulation waveforms are analyzed using SimVision. The synthesized design is further analyzed for performance parameters such as area utilization, power consumption, and timing performance.

1.2 Theory of FIR Filter

A Finite Impulse Response (FIR) filter is a digital filter whose output depends only on the current and previous input samples. Unlike other filters, FIR filters do not use feedback elements, which makes them inherently stable and easier to design.

The mathematical representation of an FIR filter is given by:

$$y[n] = h_0x[n] + h_1x[n-1] + h_2x[n-2] + \ldots + h_Nx[n-N]$$

Where:

- $x[n]$represents the input signal
- $y[n]$represents the output signal
- $h[n]$represents the filter coefficients
- N represents the number of filter taps

The FIR filter structure typically consists of delay elements, multipliers, and adders. The input signal passes through delay registers, and each delayed signal is multiplied by a corresponding coefficient. The multiplied results are then summed together to produce the final output signal.

Tap	**Operation**	**Description**
x[n]	Current input	Present input sample
x[n-1]	First delayed input	Previous input sample
x[n-2]	Second delayed input	Second previous input sample
x[n-3]	Third delayed input	Third previous input sample

Table 1.2.1: FIR Filter Tap Operations

This structure allows the filter to process signals by combining multiple delayed samples of the input signal to generate the filtered output.

1.3 Objective

The primary objective of this project is to design and simulate a Finite Impulse Response (FIR) filter using Verilog Hardware Description Language. The filter processes input samples and produces a filtered output signal based on predefined filter coefficients.

The project aims to understand the working principle of digital filtering and its hardware implementation using HDL. Functional verification of the filter is carried out using a testbench and simulation waveform analysis to ensure correct filtering operation under different input conditions.

Another objective is to perform synthesis of the design and analyze performance parameters such as area utilization, power consumption, and timing characteristics. The project provides practical exposure to digital filter design concepts and the VLSI design flow using Cadence simulation and synthesis tools.

1.4 Design Methodology

The FIR filter is implemented using behavioral modeling in **Verilog Hardware Description Language**. The design uses registers to store previous input samples and multiplies them with predefined filter coefficients.

The products obtained from the multiplication operations are summed together using adders to generate the filtered output signal. The filter operates based on a clock signal, and the input samples are processed sequentially during each clock cycle.

A testbench is developed to verify the functional behavior of the FIR filter by applying different input stimulus signals. The design is simulated using **Cadence simulation tools**, and waveform results are analyzed using SimVision to confirm correct filter operation. After successful simulation, synthesis is performed to evaluate area, power, and timing performance.

Chapter 2

DESIGN METHODOLOGY

2.1 FIR Filter Design Implementation

The digital FIR filter was implemented using Verilog Hardware Description Language. The design consists of delay elements, multipliers, adders, and filter coefficients which together form the FIR filter structure.

```
`timescale 1ns/1ps
module sync_ram (
    input clk,
    input rst, // Active high reset
    input we, // Write Enable
    input [3:0] addr, // 16 locations
    input [7:0] data_in, // 8-bit data input
    output reg [7:0] data_out // 8-bit data output
);

    // 16x8 Memory
    reg [7:0] memory [15:0];

    integer i;

    // Reset + Read/Write operation
    always @(posedge clk) begin
        if (rst) begin
            // Clear memory during reset
            for (i = 0; i < 16; i = i + 1)
                memory[i] <= 8'h00;

            data_out <= 8'h00;
        end
        else begin
            if (we) begin
                memory[addr] <= data_in; // WRITE
            end
            else begin
                data_out <= memory[addr]; // READ
            end
        end
    end

endmodule
```

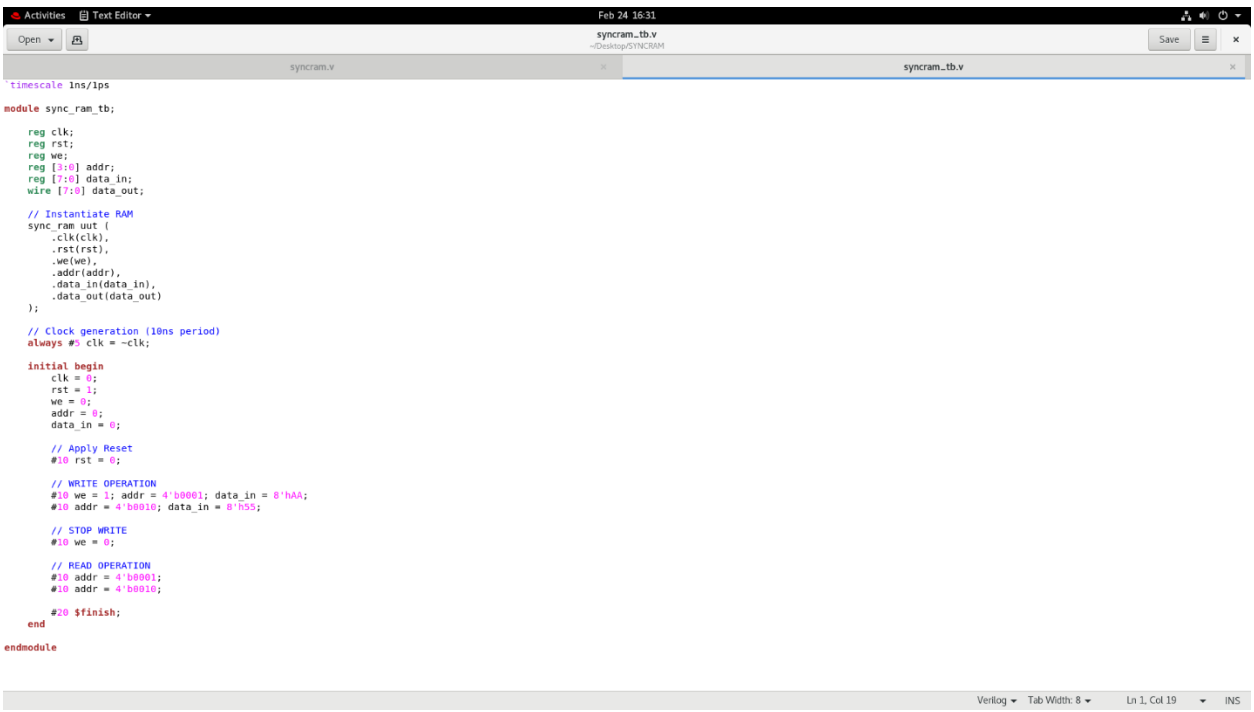
Figure 2.1.1: Verilog Implementation of FIR Filter

Figure 2.1.1 shows the Verilog implementation of the FIR filter module. The filter processes input samples sequentially and produces the output by computing the weighted sum of the current and previous input samples. The delayed input samples are multiplied by predefined filter coefficients and then summed together using

adders to generate the filtered output. Registers are used to store previous input samples so that they can be used in the filtering computation during each clock cycle.

2.2 Testbench Implementation

A testbench was developed to verify the functionality of the FIR filter design. The testbench applies clock signals and input stimulus values to observe the output behavior of the filter during simulation.



```
`timescale 1ns/1ps

module sync_ram_tb;

    reg clk;
    reg rst;
    reg we;
    reg [3:0] addr;
    reg [7:0] data_in;
    wire [7:0] data_out;

    // Instantiate RAM
    sync_ram uut (
        .clk(clk),
        .rst(rst),
        .we(we),
        .addr(addr),
        .data_in(data_in),
        .data_out(data_out)
    );

    // Clock generation (10ns period)
    always #5 clk = ~clk;

    initial begin
        clk = 0;
        rst = 1;
        we = 0;
        addr = 0;
        data_in = 0;

        // Apply Reset
        #10 rst = 0;

        // WRITE OPERATION
        #10 we = 1; addr = 4'b0001; data_in = 8'hAA;
        #10 addr = 4'b0010; data_in = 8'h55;

        // STOP WRITE
        #10 we = 0;

        // READ OPERATION
        #10 addr = 4'b0001;
        #10 addr = 4'b0010;

        #20 $finish;
    end

endmodule
```

Figure 2.2.1: Testbench Implementation of 4-Stage Pipelined Processor

Figure 2.2.1 shows the Verilog testbench used to verify the operation of the FIR filter. The testbench generates the clock signal required for sequential processing and provides different input samples to the filter. The output produced by the filter is monitored during simulation to verify correct filtering operation. The testbench ensures that the FIR filter correctly processes input samples and generates the expected output values.

2.3 RTL / Schematic View

After synthesis, the RTL schematic of the FIR filter is generated to visualize the hardware structure of the design.

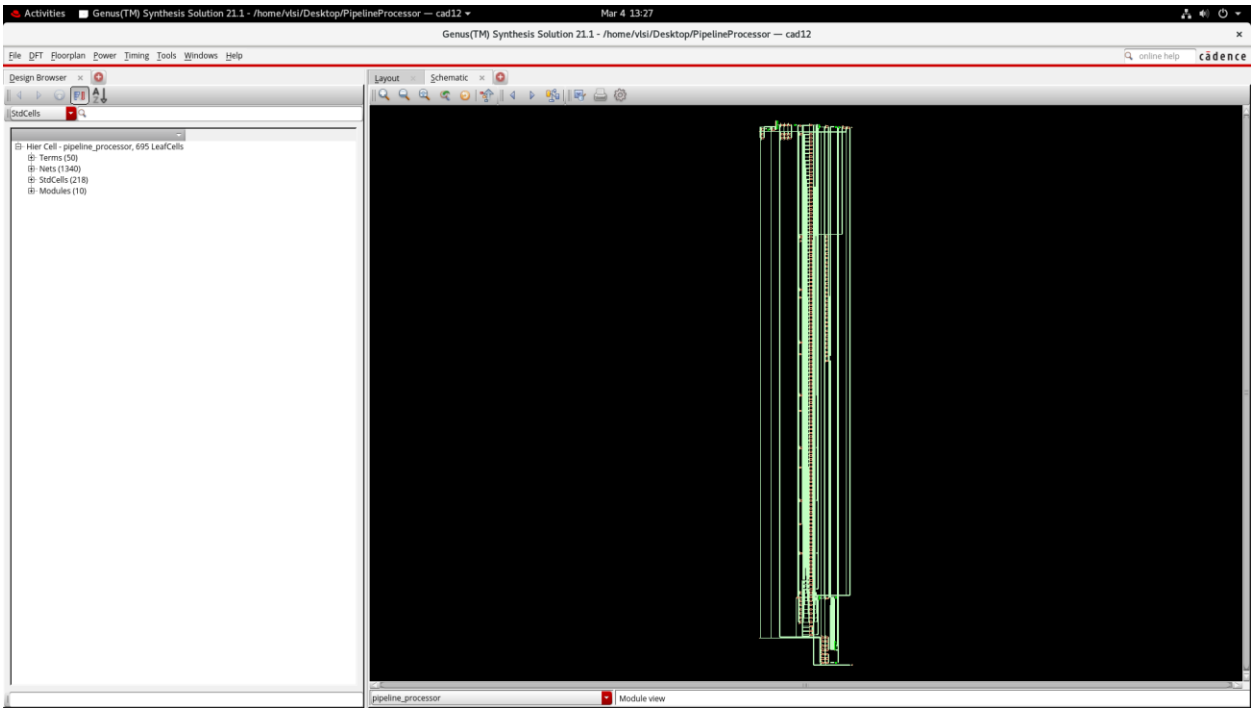


Figure 2.3.1: Synthesized RTL Schematic of FIR Filter

Figure 2.3.1 shows the synthesized RTL schematic of the FIR filter generated using Cadence tools. The schematic represents the internal hardware architecture of the filter including delay registers, multipliers, and adders used to compute the filter output. This schematic verifies that the behavioral Verilog code has been successfully converted into a hardware-level implementation suitable for digital system design.

Chapter 3

Simulation

3.1 Simulation Methodology

The FIR filter design and its testbench were simulated using **Cadence simulation tools**. The Verilog design files were compiled and executed to verify the functional behavior of the filter. The simulation was performed to ensure that the FIR filter correctly processes input samples and produces the expected filtered output during each clock cycle.

The simulation waveforms were generated using the SimVision waveform viewer. Important signals such as the clock signal, input signal, delayed input samples, filter coefficients, and output signal were observed during the simulation process. Different input values were applied through the testbench to verify the filtering operation of the design.

The results obtained from waveform analysis were compared with the expected theoretical behavior of the FIR filter to ensure correct functionality. The simulation confirmed that the filter correctly performs the weighted summation of current and previous input samples based on the predefined coefficients.

3.2 Simulation Results and Analysis

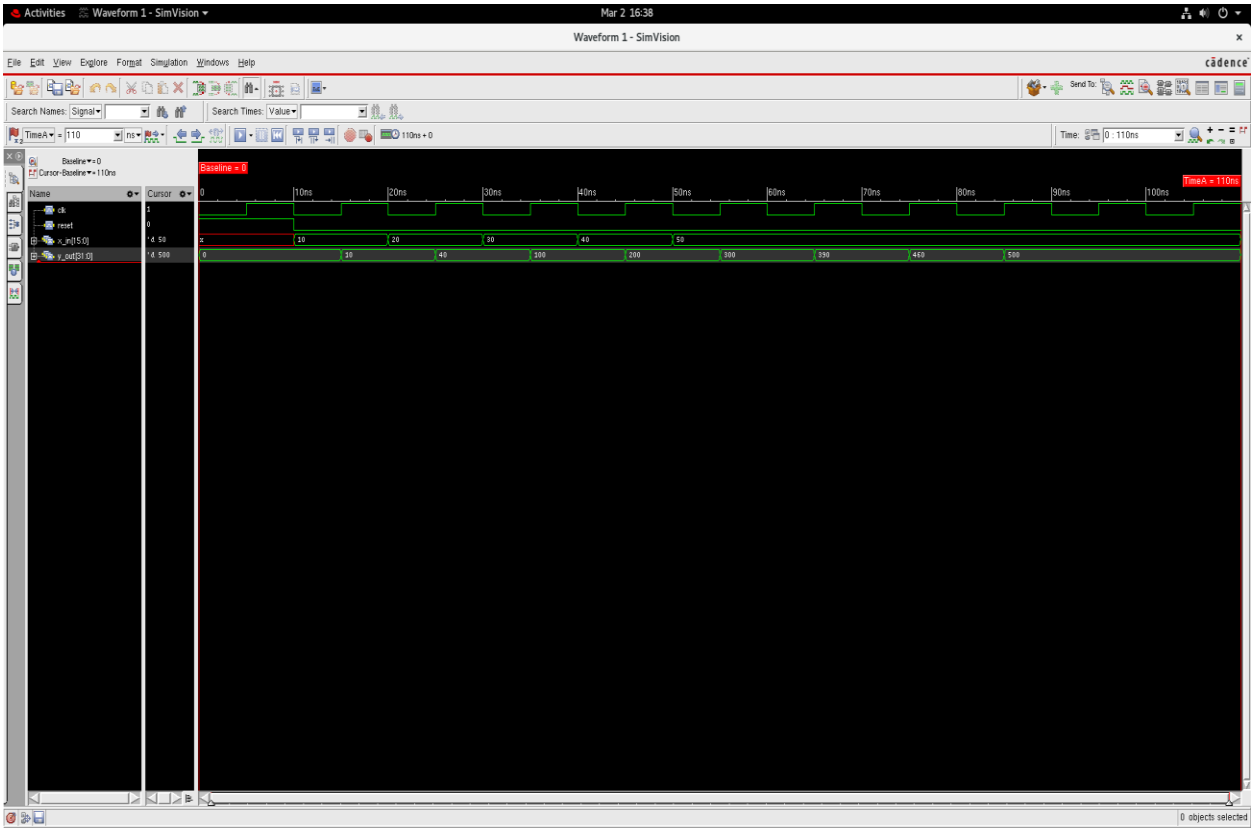


Figure 3.2.1: Simulation Waveform of FIR Filter

The simulation results of the FIR filter were verified using the SimVision waveform viewer, as shown in Figure 3.2.1. The waveform displays important signals including the clock signal, input samples, delayed input values, and the filtered output signal.

During simulation, the input signal is applied to the FIR filter and passes through a series of delay elements. Each delayed input sample is multiplied by its corresponding filter coefficient. The results of these multiplications are then summed together using adders to produce the final filtered output.

The waveform clearly shows how the input samples propagate through the delay registers and contribute to the output calculation during each clock cycle. As new

input samples are applied, previous samples shift through the delay elements, allowing the filter to compute the weighted sum of multiple input values.

The observed output waveform confirms that the FIR filter performs the filtering operation correctly according to the defined filter structure. The simulation results match the expected behavior of the designed FIR filter, verifying the functional correctness of the design.

Chapter 4

Synthesis And Performance Analysis

After successful simulation, the FIR filter design was synthesized using Cadence Genus to evaluate the hardware performance parameters of the filter implementation. Synthesis converts the behavioral Verilog description into a gate-level hardware representation suitable for physical implementation. Performance parameters such as power consumption, area utilization, and timing characteristics were analyzed using synthesis reports.

4.1 Power Analysis

The power report provides information about the dynamic and leakage power consumption of the FIR filter design.

```
Instance: /fir_filter
Power Unit: W
PDB Frames: /stim#0/frame#0
  ----------------------------------------------------------------------------
    Category         Leakage     Internal    Switching        Total    Row%
  ----------------------------------------------------------------------------
      memory     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
    register     9.26091e-06  9.41323e-04  5.04769e-04  1.45535e-03  63.71%
       latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       logic     1.02225e-05  3.53868e-04  4.18748e-04  7.82838e-04  34.27%
        bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       clock     0.00000e+00  0.00000e+00  4.61295e-05  4.61295e-05   2.02%
         pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
          pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
  ----------------------------------------------------------------------------
    Subtotal     1.94834e-05  1.29519e-03  9.69646e-04  2.28432e-03 100.00%
  Percentage           0.85%       56.70%       42.45%      100.00% 100.00%
  ----------------------------------------------------------------------------
```

Figure 4.1.1: Power Analysis Report

Figure 4.1.1 shows the power analysis report of the synthesized FIR filter generated using Cadence Genus. The report provides details about the total power consumed by the filter during operation. It includes dynamic power caused by switching

activity of the logic gates and leakage power due to static current in the circuit. Power analysis is an important parameter in VLSI design because it determines the energy efficiency of the system. The results indicate that the FIR filter operates with acceptable power consumption, demonstrating an efficient hardware implementation.

4.2 Area Analysis

The area report shows the number of logic cells and hardware resources used in the filter implementation.

```
Activities   Text Editor                    Mar 3 12:42
Open                          counter_area.rep                    Save
                              ~/Desktop/fir
============================================================
  Generated by:           Genus(TM) Synthesis Solution 21.14-s082_1
  Generated on:           Mar 02 2026  04:42:03 pm
  Module:                 fir_filter
  Technology library:     slow
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

 Instance  Module  Cell Count  Cell Area  Net Area   Total Area   Wireload
-----------------------------------------------------------------------------
fir_filter              410   3568.027     0.000     3568.027 <none> (D)
  (D) = wireload is default in technology library
```

Figure 4.2.1: Area Utilization Report

Figure 4.2.1 presents the area utilization report of the synthesized FIR filter design. The report shows the total number of standard cells and logic elements used along with the overall hardware area occupied by the circuit. Area analysis is an important factor in VLSI design because it affects chip size, fabrication cost, and hardware complexity. The synthesis results indicate that the FIR filter design uses hardware resources efficiently while maintaining correct functionality.

4.3 Timing Analysis

Timing analysis evaluates the speed and delay characteristics of the filter circuit.

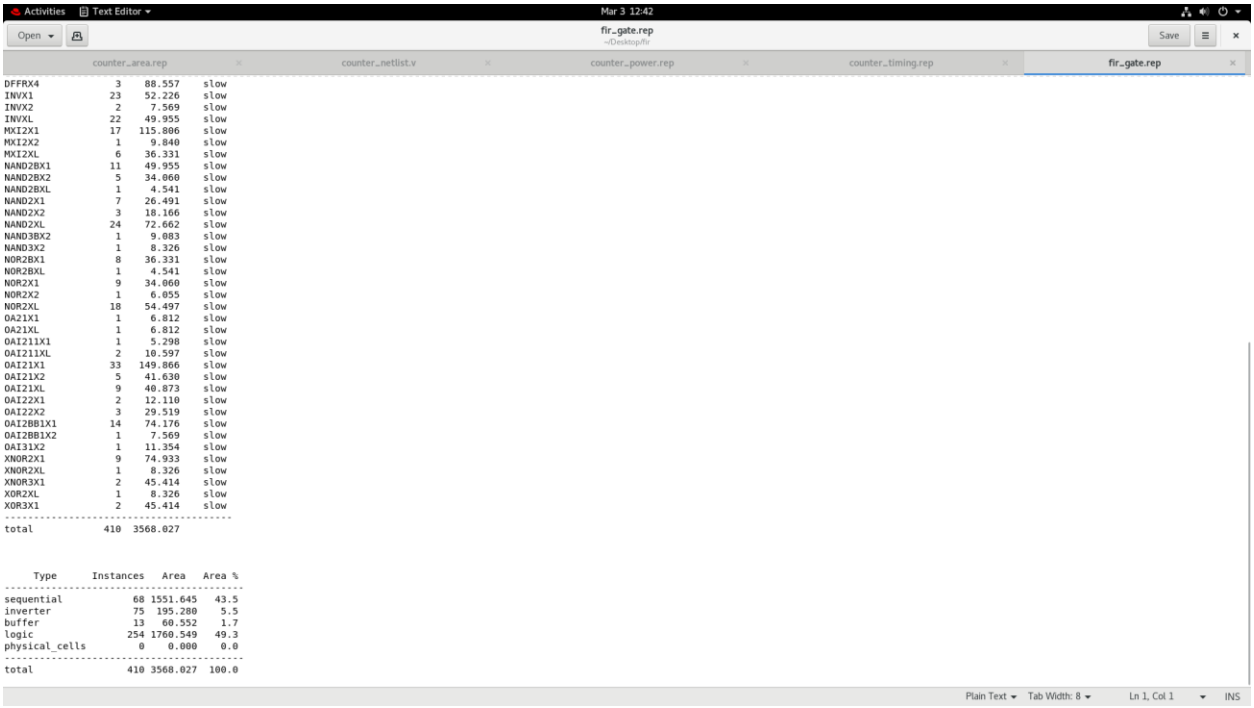
```
Activities   Text Editor                                   Mar 3 12:42
Open                                counter_timing.rep                                   Save
                                    ~/Desktop/fir
counter_area.rep   counter_netlist.v   counter_power.rep   counter_timing.rep
============================================================
  Generated by:           Genus(TM) Synthesis Solution 21.14-s082_1
  Generated on:           Mar 02 2026  04:42:03 pm
  Module:                 fir_filter
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================


Path 1: VIOLATED (-109 ps) Setup Check with Pin y_out_reg[16]/CK->D
          Group: clk
     Startpoint: (R) x2_reg[13]/CK
          Clock: (R) clk
       Endpoint: (R) y_out_reg[16]/D
          Clock: (R) clk

                     Capture       Launch
        Clock Edge:+    2000            0
      Src Latency:+       0            0
      Net Latency:+       0 (I)        0 (I)
          Arrival:=    2000            0

            Setup:-     120
      Uncertainty:-      10
     Required Time:=    1870
      Launch Clock:-       0
         Data Path:-    1978
             Slack:=    -109

#-------------------------------------------------------------------------------------------------------------
#               Timing Point                 Flags   Arc   Edge   Cell      Fanout Load Trans Delay Arrival Instance
#                                                                                (fF)  (ps)  (ps)   (ps)  Location
#-------------------------------------------------------------------------------------------------------------
  x2_reg[13]/CK                               -       -      R     (arrival)     68    -    100     0       0    (-,-)
  x2_reg[13]/Q                                -       CK->Q R      DFFRX4         3 16.6    85   429     429    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3873__1705/CO -       A->CO R      ADDFHX1        3  7.0   103   233     662    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3858/Y         -       A->Y  F      INVX1          1  2.7    47    45     707    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3818__8428/Y  -       B->Y  R      MXI2X1         1  8.6   181   156     863    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3722__2802/S  -       A->S  F      ADDFHXL        2  4.5    89   334    1197    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3961__1881/Y  -       B->Y  F      AND2X1         2  7.2    68   134    1331    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3636__9945/Y  -       A0->Y R      AOI21X2        1  5.3    78    95    1426    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3621__2802/Y  -       B0->Y F      OAI31X2        1  2.8    68    72    1498    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3604__5477/Y  -       B0->Y R      AOI21X1        1  5.3   123   111    1609    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3598__2883/Y  -       B0->Y F      OAI21X2        2  7.9    98    99    1708    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3597/Y         -       A->Y  R      CLKINVX2       2  8.1    55    63    1771    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3596/Y         -       A->Y  R      BUFX3          2  7.9    45   105    1876    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3595/Y         -       A->Y  F      CLKINVX2       1  2.6    22    26    1902    (-,-)
  csa_tree_ADD_TC_OP_4_groupi_g3584__8246/Y  -       A->Y  R      MXI2X1         1  2.2    82    76    1978    (-,-)
  y_out_reg[16]/D                            <<<     -      R     DFFRHQX8       1    -     -     0    1978    (-,-)
#-------------------------------------------------------------------------------------------------------------
```

Figure 4.3.1: Timing Analysis Report

Figure 4.3.1 shows the timing analysis report of the synthesized FIR filter design. The timing report provides information about propagation delay, critical path delay, and timing constraints of the circuit. Timing analysis ensures that the circuit operates correctly within the specified clock period without timing violations. The results confirm that the FIR filter meets the required timing constraints and operates reliably.

4.4 Gate-Level Report

The gate-level synthesis report shows the number and types of logic gates used in the filter implementation.



```
DFFRX4          3    88.557   slow
INVX1          23    52.226   slow
INVX2           2     7.569   slow
INVXL          22    49.955   slow
MXI2X1         17   115.806   slow
MXI2X2          1     9.840   slow
MXI2XL          6    36.331   slow
NAND2BX1       11    49.955   slow
NAND2BX2        5    34.060   slow
NAND2BXL        1     4.541   slow
NAND2X1         7    26.491   slow
NAND2X2         3    18.166   slow
NAND2XL        24    72.662   slow
NAND3BX2        1     9.083   slow
NAND3X2         1     8.326   slow
NOR2BX1         8    36.331   slow
NOR2BXL         1     4.541   slow
NOR2X1          9    34.060   slow
NOR2X2          1     6.055   slow
NOR2XL         18    54.497   slow
OA21X1          1     6.812   slow
OA21XL          1     6.812   slow
OAI211X1        1     5.298   slow
OAI211XL        2    10.597   slow
OAI21X1        33   149.866   slow
OAI21X2         5    41.630   slow
OAI21XL         9    40.873   slow
OAI22X1         2    12.110   slow
OAI22X2         3    29.519   slow
OAI2BB1X1      14    74.176   slow
OAI2BB1X2       1     7.569   slow
OAI31X2         1    11.354   slow
XNOR2X1         9    74.933   slow
XNOR2XL         1     8.326   slow
XNOR3X1         2    45.414   slow
XOR2XL          1     8.326   slow
XOR3X1          2    45.414   slow
-------------------------------------
total         410  3568.027


     Type      Instances   Area   Area %
-------------------------------------------
sequential            68 1551.645   43.5
inverter              75  195.280    5.5
buffer                13   60.552    1.7
logic                254 1760.549   49.3
physical_cells         0    0.000    0.0
-------------------------------------------
total                410 3568.027  100.0
```

Figure 4.4.1: Gate-Level Synthesis Report

Figure 4.4.1 shows the gate-level synthesis report generated after synthesizing the FIR filter design. The report provides information about the different types of logic gates used in the hardware implementation of the filter. Gate-level analysis helps in understanding the hardware complexity and structural efficiency of the design. The results indicate that the FIR filter is implemented using an optimized number of logic gates, ensuring efficient hardware realization.

Chapter 5

Results

The simulation results confirm the correct functional operation of the designed FIR filter for all tested conditions. During simulation, input samples were applied to the filter through the testbench, and the filter successfully processed the input signals using delay elements, multipliers, and adders to generate the filtered output.

The waveform analysis clearly demonstrated that the input samples passed through the delay registers, allowing the filter to compute the weighted sum of current and previous input samples based on the predefined filter coefficients. As new input values were applied during each clock cycle, the previous input samples shifted through the delay elements, enabling the filter to perform the required filtering operation.

Furthermore, synthesis analysis provided important information about the hardware performance of the FIR filter design. The area report indicated efficient utilization of logic cells and hardware resources, demonstrating an optimized hardware implementation of the filter structure. The power analysis showed acceptable dynamic and leakage power consumption, indicating energy-efficient circuit operation. Timing analysis verified that the design meets the required timing constraints without violations, ensuring reliable and stable filter operation.

Overall, the obtained results validate both the functional correctness and hardware efficiency of the designed FIR filter.

Chapter 6

Discussion

This project demonstrates the implementation of a Finite Impulse Response (FIR) filter using Verilog Hardware Description Language. The filter design was modeled and verified through simulation using Cadence tools. The design illustrates how digital filters process discrete input samples and generate filtered output signals using delay elements, multipliers, and adders.

The FIR filter operates by computing the weighted sum of the current and previous input samples based on predefined filter coefficients. Delay registers are used to store previous input samples, allowing the filter to process multiple input values during each clock cycle. This structure ensures stable and reliable filter operation since FIR filters do not use feedback elements.

The functional verification through simulation confirmed that the FIR filter correctly processes the input samples and produces the expected filtered output. The waveform analysis clearly showed the movement of input samples through the delay elements and the generation of the output signal through multiplication and accumulation operations.

The synthesis results further confirmed that the FIR filter design is hardware-efficient and meets important performance parameters such as area utilization, power consumption, and timing constraints. Overall, this project demonstrates the practical implementation of a digital filter and the VLSI design flow using Verilog and Cadence simulation tools.

Chapter 7

Conclusion

The FIR (Finite Impulse Response) filter was successfully designed, implemented, simulated, and synthesized using Verilog Hardware Description Language and Cadence tools. The filter processes input samples and produces filtered output by performing weighted summation of current and previous input values using predefined filter coefficients.

Functional verification through simulation confirmed that the FIR filter correctly performs the filtering operation for different input samples. The simulation waveform demonstrated the proper operation of delay elements, multipliers, and adders used in the filter structure.

The synthesis analysis showed efficient utilization of hardware resources with acceptable area, power consumption, and timing performance. The design successfully meets the required functionality and performance requirements.

Overall, this project provided practical experience in digital filter design, simulation, and synthesis, and demonstrated the implementation of FIR filters using the VLSI design flow.